



Antenna Specifications

WNC PN: 81VPAG15.G01

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Revision History

Rev. #	Author	Summary of Changes	Date
01		New release	2024/06/24
02		Update Outdoor Specification	2024/08/16
03		Update IP67 Antenna Specification	2025/04/18
04		Update Antenna data	2026/02/10

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1. Introduction

This specification covers the FR4 antenna for sub6G and NTN.

2. Electrical Specifications

The antenna has the electrical characteristics given in Table 1 under WNC standard installation conditions shown in the figure.

Electrical Characteristics for sub6G					
Freq. (MHz)	617~960	1450~1605	1710~2690	3300~4900	5150~5925
V.S.W.R.	≤ 2.5	≤ 2.5	≤ 2.3	≤ 2.5	≤ 3.0
Peak Gain	2.5 dBi	1.5 dBi	3.9 dBi	3.7 dBi	4.2 dBi
Eff. % (max./avg.)	51/43	54/51	80/69	70/61	68/58
Impedance	50 Ohm				
Connector	SMA				
Substrate	FR4				
Dimension	172.5 mm x 27.9mm				
IP Rating	IP67				

Electrical Characteristics for NTN			
Band	n23	n255	n256
Freq. (MHz)	2000~2200	1525~1660.5	1980~2200
V.S.W.R.	≤ 2.0	≤ 2.5	≤ 2.0
Peak Gain	3.0 dBi	1.2 dBi	3.0 dBi
Eff. % (max./avg.)	75/69	52/51	75/68
Impedance	50 Ohm		
Connector	SMA		
Substrate	FR4		
Dimension	172.5 mm x 27.9mm		
IP Rating	IP67		

3. Environmental conditions

3.1 Operating conditions

The antenna has the electrical characteristics given in Tables 1 in the temperature range of - 20°C to +65°C.

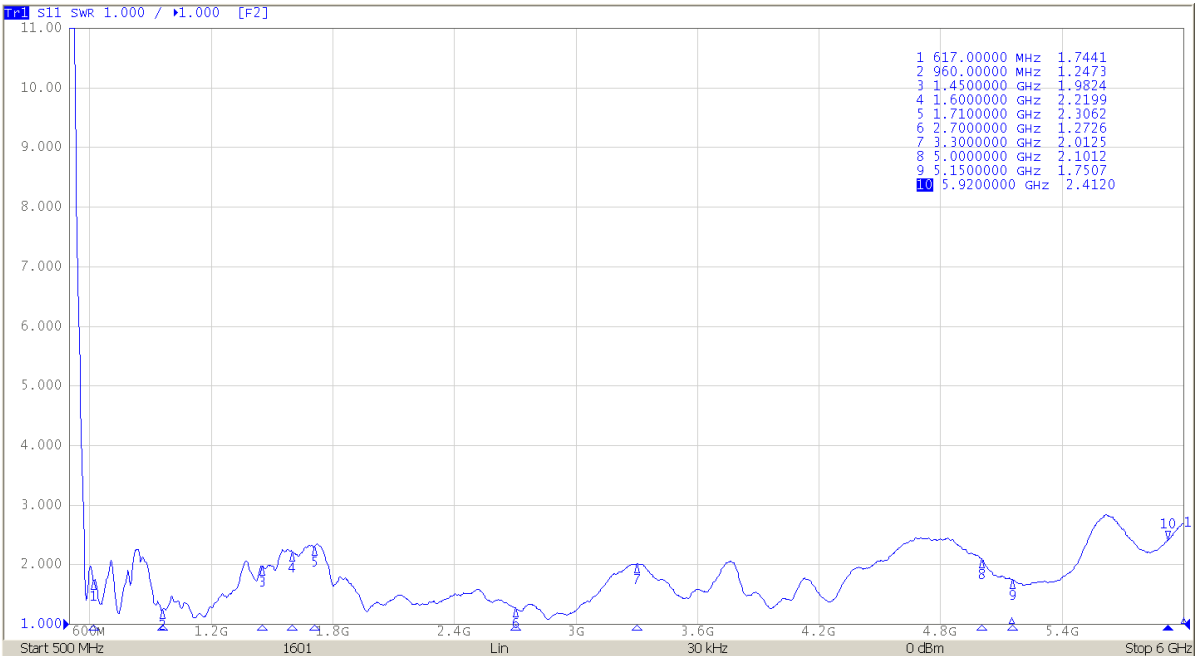
3.2 Storage temperature range

The storage temperature range of product is -40°C to +70°C.

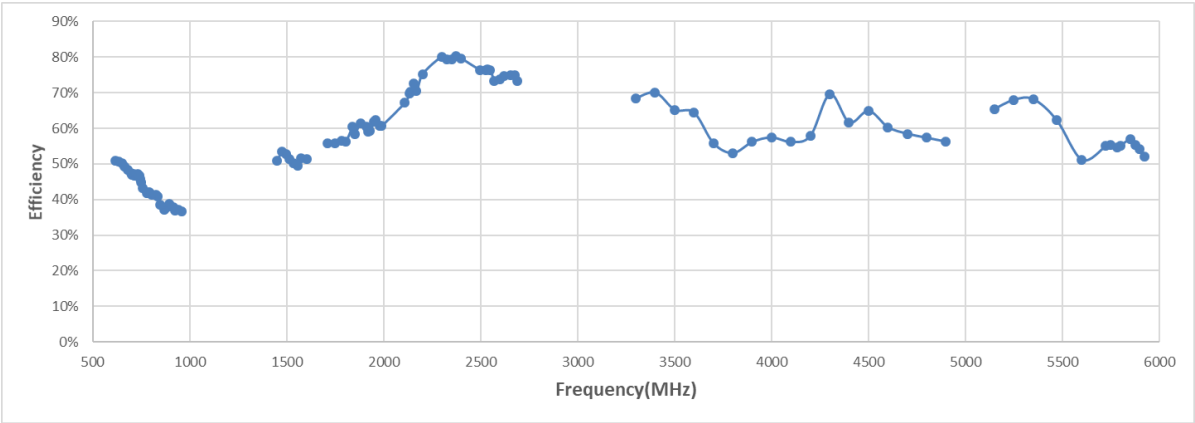
4. Shape and Dimension



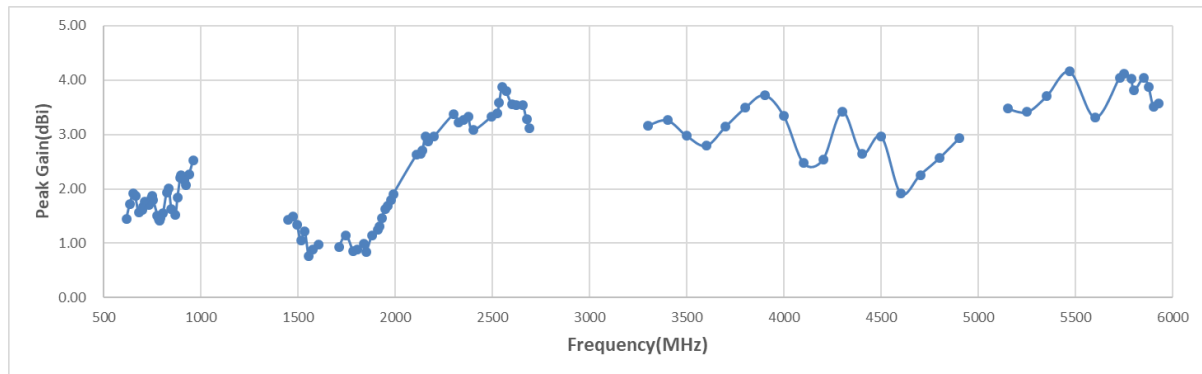
5. VSWR



6. Efficiency



7. Peak Gain



8. Raw Data

Frequency (MHz)	Efficiency	Avg. Gain	Peak Gain
617	51.0%	-2.93	1.45
634	50.6%	-2.96	1.72
652	50.1%	-3.00	1.91
663	49.2%	-3.08	1.87
680	48.2%	-3.17	1.57
699	47.4%	-3.25	1.62
704	47.0%	-3.27	1.68
710	47.1%	-3.27	1.77
716	46.7%	-3.31	1.75
734	47.2%	-3.26	1.70
740	46.7%	-3.31	1.75
746	45.7%	-3.40	1.83
751	44.9%	-3.48	1.87
756	43.3%	-3.64	1.80
777	41.7%	-3.80	1.51
782	41.8%	-3.78	1.45
787	42.0%	-3.77	1.42
791	42.1%	-3.76	1.47
806	41.3%	-3.84	1.56
824	41.3%	-3.84	1.94
836	40.8%	-3.89	2.01
849	38.4%	-4.15	1.63
869	37.1%	-4.31	1.53
880	37.7%	-4.23	1.84
894	38.9%	-4.10	2.21
900	38.4%	-4.16	2.25
915	37.9%	-4.21	2.16
925	36.8%	-4.34	2.07
940	37.0%	-4.32	2.27
960	36.6%	-4.37	2.52

Frequency (MHz)	Efficiency	Avg. Gain	Peak Gain
1450	50.9%	-2.93	1.43
1475	53.6%	-2.71	1.50
1495	52.9%	-2.77	1.34
1515	51.4%	-2.89	1.05
1535	50.2%	-2.99	1.22
1555	49.5%	-3.05	0.77
1575	51.7%	-2.87	0.88
1605	51.3%	-2.90	0.98
1710	55.9%	-2.53	0.93
1747	55.9%	-2.53	1.15
1785	56.5%	-2.48	0.85
1805	56.4%	-2.49	0.89
1840	60.4%	-2.19	1.00
1850	58.5%	-2.33	0.84
1880	61.3%	-2.12	1.14
1910	60.4%	-2.19	1.26
1920	59.0%	-2.29	1.31
1930	59.3%	-2.27	1.46
1950	61.9%	-2.08	1.64
1960	62.3%	-2.06	1.69
1980	60.8%	-2.16	1.80
1990	60.7%	-2.17	1.91
2110	67.2%	-1.72	2.63
2132	69.7%	-1.57	2.65
2140	70.2%	-1.54	2.71
2155	72.6%	-1.39	2.97
2170	70.5%	-1.52	2.88
2200	75.2%	-1.24	2.97
2300	80.2%	-0.97	3.37
2325	79.5%	-1.00	3.22
2350	79.3%	-1.01	3.27
2375	80.3%	-0.95	3.33
2400	79.6%	-0.99	3.09
2496	76.4%	-1.17	3.33
2525	76.3%	-1.18	3.38
2535	76.5%	-1.17	3.59
2550	76.3%	-1.18	3.88
2570	73.3%	-1.35	3.80
2600	73.7%	-1.33	3.55
2620	74.7%	-1.27	3.54
2655	75.0%	-1.25	3.54
2675	74.8%	-1.26	3.29
2690	73.2%	-1.36	3.12

Frequency (MHz)	Efficiency	Avg. Gain	Peak Gain
3300	68.4%	-1.65	3.17
3400	70.0%	-1.55	3.27
3500	65.2%	-1.86	2.98
3600	64.4%	-1.91	2.80
3700	55.8%	-2.53	3.15
3800	53.1%	-2.75	3.49
3900	56.2%	-2.51	3.73
4000	57.4%	-2.41	3.34
4100	56.2%	-2.51	2.48
4200	57.9%	-2.37	2.54
4300	69.6%	-1.58	3.42
4400	61.7%	-2.10	2.65
4500	64.9%	-1.88	2.97
4600	60.3%	-2.20	1.92
4700	58.5%	-2.33	2.26
4800	57.3%	-2.42	2.57
4900	56.3%	-2.50	2.93
5150	65.4%	-1.84	3.48
5250	68.0%	-1.67	3.42
5350	68.1%	-1.67	3.70
5470	62.3%	-2.06	4.16
5600	51.2%	-2.91	3.31
5725	55.1%	-2.59	4.04
5750	55.4%	-2.56	4.11
5785	54.7%	-2.62	4.03
5800	55.2%	-2.58	3.81
5850	57.0%	-2.44	4.05
5875	55.4%	-2.57	3.88
5900	54.3%	-2.65	3.52
5925	52.1%	-2.83	3.57

9. Radiation Pattern

